Daffodil International University
Department of CSE
CSE 411 – Artificial Intelligence
Class Test

Time: 30 Minutes **Total Marks:** 15

Answer all questions. Each question carries 5 marks.

Question 1 (6 Marks) – Perceptron Learning Rule

A single-layer perceptron has the following parameters:

• Learning rate, $\eta = 0.1$
• Input vector, $X = (1,0,1)$
• Initial weight vector, $W = (0.2,0.4,0.3)$
• Bias, $b = -0.6$
• Target output, $t = 1$

Using the step activation function:
$f(net)=1$ if $net \geq 0$, otherwise 0.

a) Calculate the net input:
$net = \Sigma(x_i w_i)+b$
b) Determine the actual output of the perceptron. Calculate the error $(t-y)$.
c) Update the weights using the Perceptron Learning Rule and Write the updated weight vector.

Question 2 (3 Marks) – Network Architecture

Difference Between Feedforward Neural Network vs Feedback (Recurrent) Neural Network.

Question 3 (6 Marks) – Bayes' Theorem

Consider a football game between two rival teams, Team 0 and Team 1.

• Team 0 wins 65% of all matches.
• Team 1 wins the remaining matches.
• Among the matches won by Team 0, 30% are played at Team 1's home ground.
• Among the matches won by Team 1, 75% are played at Team 1's home ground.

If Team 1 hosts the next match, use Bayes' Theorem to determine which team is more likely to win. Show all necessary calculations.